

Software Design Document (Frontend)

1. Introduction

This document details the architectural and structural design of the Gamified Platform's frontend application.

2. System Overview

The frontend serves as the interactive layer for learners and admins. It handles authentication, curriculum navigation, gamification visualization, and AI tutoring interaction.

3. Technology Stack

Framework: Next.js 14 (App Router)

Language: JavaScript (ES6+)

Styling: Tailwind CSS

State Management: React Context API

API Client: Vanilla fetch with lib/api.js interceptor

4. Subsystem Design

4.1 Authentication: Custom useRequireAuth hook for session persistence and role-based gating.

4.2 Gamification: Real-time XP/Coin synchronization with backend wallet.

4.3 API Layer: lib/api.js handles 401 Unauthorized by attempting a background refresh of JWT tokens.

5. Core React Patterns

Layouts: Shared layouts for student and admin hubs.

Hooks: useRequireAuth, custom data fetching hooks.

Components: Atomic design for reusable UI widgets.

6. API Communication

Consistent success/failure handling mirroring the backend JSON structure. Implements robust error reporting for Zod-validated failures.

7. Security

Client-side RBAC (Learner vs Admin vs SuperAdmin).

Sensitive routes protected via middleware redirect logic.

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Token storage in LocalStorage (mirroring secure cookie behavior for development).